

SHP ASSET MANAGER

LAZER EYE

User Manual



scotthendersonphotography.net



CONTENTS

01	System Overview (#section-1)	02	Connecting to the App (#section-2)
03	433 MHz Receiver Binding (#section-3)	04	Configuration Settings (#section-4)
05	Laser Alignment (#section-5)	06	Wake Trigger (#section-6)
07	Advanced Settings (#section-7)	08	ESP-NOW Wireless Receiver (#section-8)
09	Status LEDs & Battery (#section-9)	10	Troubleshooting (#section-10)

SYSTEM OVERVIEW

The **Lazer Eye** is a LiDAR-based camera trigger. It continuously measures the distance to whatever is in front of it and fires your camera shutter the moment something enters a distance window you define – no button press required.

Common uses include wildlife photography (animal crosses the beam), action sports (athlete reaches a point), product photography, and long-exposure light painting.

TWO-PART SYSTEM

LAZER EYE

The main sensor unit. Measures distance, connects to the SHP Asset Manager app over Bluetooth or WiFi, and fires your camera when a subject enters the zone you have set.

ESP32 MINI (OPTIONAL)

A small wireless receiver that sits at the camera end. Receives trigger and wake signals from the Lazer Eye over radio – eliminating the need for a cable run between the sensor and your camera.

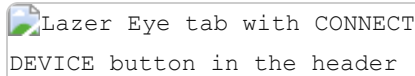
TWO TRIGGER OUTPUTS

The Lazer Eye has two independent outputs that both fire at the same instant. You can use one or both simultaneously.

OUTPUT	WHAT IT DOES	BEST FOR
433 MHz (always active)	Drives a 433 MHz RF receiver at the camera end. The receiver plugs into your camera's 3.5 mm shutter jack.	Simple setup, zero-latency, works straight out of the box.
ESP-NOW wireless (optional)	Sends trigger and wake signals wirelessly to a paired ESP32 Mini. No cable required between the Lazer Eye and the camera.	Remote or awkward placements, water environments, moving rigs.

CONNECTING TO THE APP

The SHP Asset Manager app controls all Lazer Eye settings in real time. Connect via **Bluetooth** or **WiFi** – both give full control.



Tap ⚡ **CONNECT DEVICE** to start pairing

BLUETOOTH

- 1 Power on the Lazer Eye.
- 2 Ensure Bluetooth is enabled on your phone.
- 3 Open the app and tap the **LAZER EYE** tab at the bottom of the screen.
- 4 Tap ⚡ **CONNECT DEVICE** in the header and select your device from the scan LASER_EYE list. The default device name is .
- 5 Settings sync automatically. The dot in the header turns orange when connected.

The app remembers your device. On subsequent launches it reconnects silently if the Lazer Eye is in range.

WIFI

The Lazer Eye creates its own WiFi hotspot. WiFi is recommended when installing firmware updates – it is significantly faster than Bluetooth.

- 1 Power on the Lazer Eye.
- 2 On your LASER_EYE network in OS WiFi settings. If you have set a Security PIN of 8 or more characters in the app, that PIN is also the join the hotspot password.
- 3 Return to the app. Tap ⚡ **CONNECT DEVICE** → *Connect via WiFi* .
- 4 Enter IP **192.168.4.1** and port **80** , then tap CONNECT.
- 5 Settings sync automatically once connected.

433 MHZ RECEIVER BINDING

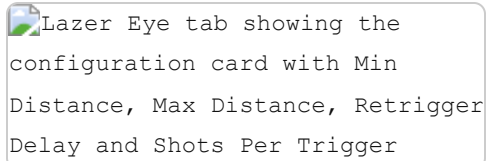
The 433 MHz receiver at your camera end must be *bound* to the Lazer Eye before it will relay trigger signals. This is a one-time step. The receiver plugs into your camera's 3.5 mm shutter jack.

- 1 Power on the 433 MHz receiver at the camera end.
- 2 Locate **SYNC** or **LEARN** button on the receiver and hold it until it enters pairing the mode (typically 2-5 seconds).
- 3 Power on the Lazer Eye. It transmits a 433 MHz signal automatically on boot.
- 4 The receiver confirms the binding. Pairing is stored in the receiver's memory and does not need to be repeated unless you replace either unit.

Once bound, every trigger and wake event automatically fires the 433 MHz output – no setting in the app controls this. It is always active.

CONFIGURATION SETTINGS

The **CONFIGURATION** card is expanded by default at the top of the Lazer Eye tab. These four values define the trigger zone and burst behaviour.



Configuration card – Min/Max Distance, Retrigger Delay, Shots Per Trigger

MIN DISTANCE

0 cm up to Max Distance

The closest range at which the Lazer Eye will trigger. Objects nearer than this are ignored entirely.

MAX DISTANCE

Min Distance up to 800 cm

The furthest range that triggers a shot. Objects beyond this are ignored. The orange zone on the distance bar shows your active trigger window.

RETRIGGER DELAY

0.0 s and above

Minimum time between successive triggers. Set to **0.0 s** to fire on every sensor reading. Increase to limit how rapidly the camera fires when a subject lingers in the zone.

SHOTS PER TRIGGER

1 and above

How many photos are taken each time the trigger fires. **1** for single capture; increase for burst sequences.

All values are written to the device immediately when changed and are retained after power-off.

READING THE DISTANCE MONITOR

- A The large number at the top shows the live distance reading.
- A The **orange zone** on the bar marks your Min-Max trigger window. When the moving dot enters it the shutter fires and the display flashes **TRIGGER**.
- A Tap **CM/M** or **FT/IN** to switch the display unit.
- A The **AVG xN** badge shows the current smoothing level (adjustable in Advanced).

LASER ALIGNMENT

The Lazer Eye has a built-in alignment laser that projects a visible dot at exactly the point the sensor is measuring. Use it before a shoot to confirm the sensor is aimed at the right spot in the frame.

☐A Toggle **LASER ALIGNMENT** in the Lazer Eye tab to turn the dot on.

☐A The laser stays on for **5 minutes** then shuts off automatically. Tap the toggle again to turn it off early.

☐A The alignment laser does not affect trigger behaviour – the device continues to detect and fire normally while it is on.

Do not stare directly into the alignment laser beam.

WAKE TRIGGER

Many cameras enter a sleep or standby mode after a period of inactivity and miss the first shutter trigger after a long wait. Wake Trigger sends a short periodic pulse to keep your camera awake between triggers.

- 1 Scroll to the **WAKE TRIGGER** section in the Lazer Eye tab.
- 2 Toggle the switch **ON** . The label turns orange when active.
- 3 Select a preset interval or enter a custom value.

PRESET	INTERVAL	RECOMMENDED FOR
30s	30 seconds	Cameras with very aggressive sleep settings
1 min	60 seconds	General use – most cameras
2 min	120 seconds	Cameras with 1-2 min auto-off
5 min	300 seconds	Long waits – wildlife setups
10 min	600 seconds	Very long waits
30 min	1800 seconds	Overnight setups
CUSTOM	1-1800 s	Any specific interval

When connected, a **NEXT PULSE IN** countdown appears at the bottom of the card showing when the next wake pulse will fire.

Set the wake interval slightly shorter than your camera's auto-off time. If the camera sleeps after 1 minute, use a 45-second interval.

ADVANCED SETTINGS

Tap the **ADVANCED** row to expand these controls. Most users can leave them at their defaults.



Full Lazer Eye tab showing the
ADVANCED row collapsed at the
bottom of the screen

ADVANCED row – tap to expand

SCAN RATE (HZ)

Default: 100 Hz

How many distance readings per second the sensor takes. Higher values make the system more responsive to fast-moving subjects but draw more power.

PULSE SPEED (MS)

Milliseconds

How long the trigger signal is held active on each fire. Increase if your camera consistently misses triggers – some bodies need a longer pulse to register reliably.

TRIGGER INVERSE

Toggle – default OFF

Flips the trigger polarity. Enable if your camera fires *between* subjects instead of *on* them.

SMOOTHING

1 (off) to 10

Averages distance readings over a rolling window. Higher values reduce false triggers from noise at long range, at the cost of a small amount of added latency.

STATUS LED

Toggle – default ON

The onboard LED blinks on every trigger and wake event. Disable in environments where stray light would affect the shot or disturb wildlife.

POWER LED

Toggle – default ON

Solid light that shows the device is powered on. No trigger correlation. Disable for dark-environment or long-exposure work.

MIN FLUX – SIGNAL QUALITY GATE

The LiDAR sensor returns a **signal strength value** (flux) alongside every distance reading. Readings with very low signal strength – caused by glancing angles off shiny surfaces or the beam catching an edge – can produce false distance values that trigger the camera incorrectly.

MIN FLUX silently discards any reading below the threshold. Only readings that pass the gate are used.

VALUE	EFFECT	WHEN TO USE
0	No filtering	Very close range, matte subjects only
20 (default)	Rejects most phantom returns	General purpose – start here
50–100	Strict filtering	Shiny, metallic, or distant subjects causing false triggers
200+	Very strict – may miss weak returns	High-glare environments only

If dark or matte subjects (e.g. black clothing) stop being detected, reduce Min Flux. If shiny subjects cause false triggers, increase it.

ESP-NOW WIRELESS RECEIVER

The ESP32 Mini is an optional wireless receiver that sits at the camera end. It receives trigger and wake signals from the Lazer Eye over a direct radio link – no WiFi router required. Pairing takes about two minutes.



ESP-NOW card – requires live device connection

PAIRING THE ESP32 MINI

- 1 Connect to your Lazer Eye in the app via Bluetooth or WiFi.
- 2 Scroll to the **ESP-NOW WIRELESS RECEIVER** card and toggle it **ON**.
- 3 Tap  icon **RECEIVER MAC**. The app reads and displays the Lazer Eye's wireless ID. Tap the **refresh** next to **ADDRESS**.
- 4 Obtain the ESP32 Mini's MAC address – it is printed to the Serial Monitor at 115 200 baud when the Mini first boots after being flashed.
- 5 Tap **PAIR** field and enter the Mini's MAC address AA:BB:CC:DD:EE:FF. Tap the **TRANSMITTER MAC** in the format
- 6 Tap **PAIR**. The Lazer Eye registers the Mini and begins sending **TRANSMITTER** packets.
- 7 The **PAIRED** badge turns green as soon as the Mini receives its first packet. The Mini's battery percentage also appears in the app header.

Pairing survives reboots on both devices. You only need to repeat this if you replace either unit or want to pair to a different Mini.

RESETTING THE MINI'S PAIRING

- 1 Power off the ESP32 Mini.
- 2 Hold **BOOT** and apply power, continuing to hold **3**, then the **button** for **seconds** release.

3

The Mini clears its stored pairing and slow-blinks its LED, indicating it is waiting for a new first packet.

MAC ADDRESS — WHICH IS WHICH?

FIELD IN APP	WHOSE ADDRESS	WHERE TO GET IT
RECEIVER MAC ADDRESS	The Lazer Eye's wireless ID	Tap  refresh in the app while connected
PAIR TRANSMITTER MAC	The ESP32 Mini's address — entered once during setup	Serial Monitor at 115 200 baud on first boot

STATUS LEDS & BATTERY

LAZER EYE LEDS

LED	BEHAVIOUR	MEANING
Status LED	Single blink	A trigger or wake pulse just fired. Can be disabled in Advanced → Status LED.
Power LED	Solid on	Device is powered on and running. Can be disabled in Advanced → Power LED.

ESP32 MINI LED

LED	BEHAVIOUR	MEANING
Status LED	Solid on	Paired to Lazer Eye – ready to fire.
Status LED	Slow blink (~1 Hz)	Not yet paired – waiting for first wireless packet.
Status LED	Single short flash	Trigger received – camera shutter fired.
Status LED	Double flash	Wake received – camera keep-alive pulse fired.

BATTERY

The battery percentage for the Lazer Eye appears as a battery icon in the app header when connected. For the ESP32 Mini, the percentage is reported back to the app every 30 seconds and shown as *TX Battery* in the header.

Charge both devices via their **USB-C ports** using any standard 5 V USB charger. The charge IC manages the charge profile automatically.

TROUBLESHOOTING

False triggers when nothing is in frame	Increase Min Flux in Advanced – start at 50 and increase until false triggers stop. The sensor is catching low-confidence reflections from a background or shiny surface.
Sensor stops detecting subjects	Reduce Min Flux . It may be set too strict for dark or matte subjects (e.g. black clothing). Also confirm the Min-Max trigger zone covers the subject's position using the live distance monitor.
Camera misses the first shot after a long wait	Enable Wake Trigger at an interval slightly shorter than your camera's auto-off time. If the camera sleeps after 1 minute, use a 45-second interval.
Camera fires between subjects, not on them	Enable Trigger Inverse in Advanced settings.
Camera misses short triggers	Increase Pulse Speed in Advanced. Some camera bodies require a longer signal to reliably register a shutter command.
ESP-NOW shows NOT PAIRED	Ensure the Mini is powered on. Re-enter its MAC address and tap PAIR TRANSMITTER again. Confirm ESP-NOW toggle is ON and the Lazer Eye is connected. A slow-blinking LED on the Mini means it is waiting for its first packet.
433 MHz receiver not firing	Re-bind the receiver: hold SYNC / LEARN until it enters pairing mode, then power-cycle the Lazer Eye.
Can't connect via Bluetooth	Toggle Bluetooth off and on and power-cycle the Lazer Eye. On iOS: Settings → Privacy → Bluetooth → confirm the SHP Asset Manager app has permission.
Firmware update is slow or failing	Switch to a WiFi connection – OTA updates over WiFi are dramatically faster than Bluetooth. Settings → Hardware Settings → Check for Updates.

UPDATING FIRMWARE

- 1 Connect to the Lazer Eye in the app.

- 2 Tap the  **gear icon** in the header to open Settings.
- 3 Scroll to **Hardware Settings** and tap **CHECK FOR UPDATES** .
- 4 If an update is available tap **INSTALL UPDATE** . The device reboots and the app reconnects automatically.